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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	No debug kaw kaw
Selected Domain	Domain 1
Selected SDG Track (1–17)	4/8/9
Project / Solution Name	JobBud
Name (Leader)	Po Rou Jing
Email & Phone (Leader)	roujing2005@gmail.com +6016-294 3869
University Affiliation(s)	Universiti Malaya
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Fresh graduates and career switchers navigating analytical, financial, and technical fields face a critical challenge: aligning their existing capabilities with rapidly changing industry requirements without clear, data-driven guidance. As automation, artificial intelligence, and data-intensive roles reshape career pathways, job seekers may struggle to assess their current skill readiness, identify transferable strengths, and understand specific competency gaps in areas such as quantitative analysis, complex problem-solving, systems thinking, and digital tools.

This challenge has broader economic and social implications. Limited access to transparent career guidance can contribute to career mismatch, underemployment, and inefficient workforce transitions, affecting productive employment under Sustainable Development Goal (SDG) 8 and lifelong skills development under SDG 4. TalentCorp's workforce impact study further highlights the scale of workforce transformation in Malaysia, showing that technological and economic shifts are expected to significantly affect a substantial number of workers.

JobBud addresses this information gap by comparing user-provided skills and knowledge with structured occupational evidence, identifying development priorities, and presenting personalised reskilling pathways. The solution aims to support more informed career decisions while contributing to skills development, decent work, and innovation under SDGs 4, 8, and 9.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

JobBud is an intelligent web-based career guidance solution designed to help fresh graduates and career switchers navigate rapidly evolving job markets through transparent, evidence-based recommendations.

The application uses deterministic computational methods to compare user-provided skills, knowledge, and self-reported technology experience with structured occupational data from O*NET. It produces explainable career-alignment scores, identifies strengths and development priorities, and generates personalised reskilling pathways using verified or team-curated learning resources.

JobBud also incorporates updatable labour-market and technology signals from public sources, including current job postings and AI-related news attention, to provide additional market context. These signals are presented as indicators rather than guaranteed forecasts. Moderate and fast AI-disruption scenarios allow users to explore how career comparisons may change under different assumptions.

The prototype is developed in Python using Streamlit and deployed through Replit. An optional RunPod AI layer is designed to convert already-computed structured results into concise natural-language explanations without determining or modifying career scores.

Career recommendations remain informational guidance, with human judgement kept at the centre of career decisions.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

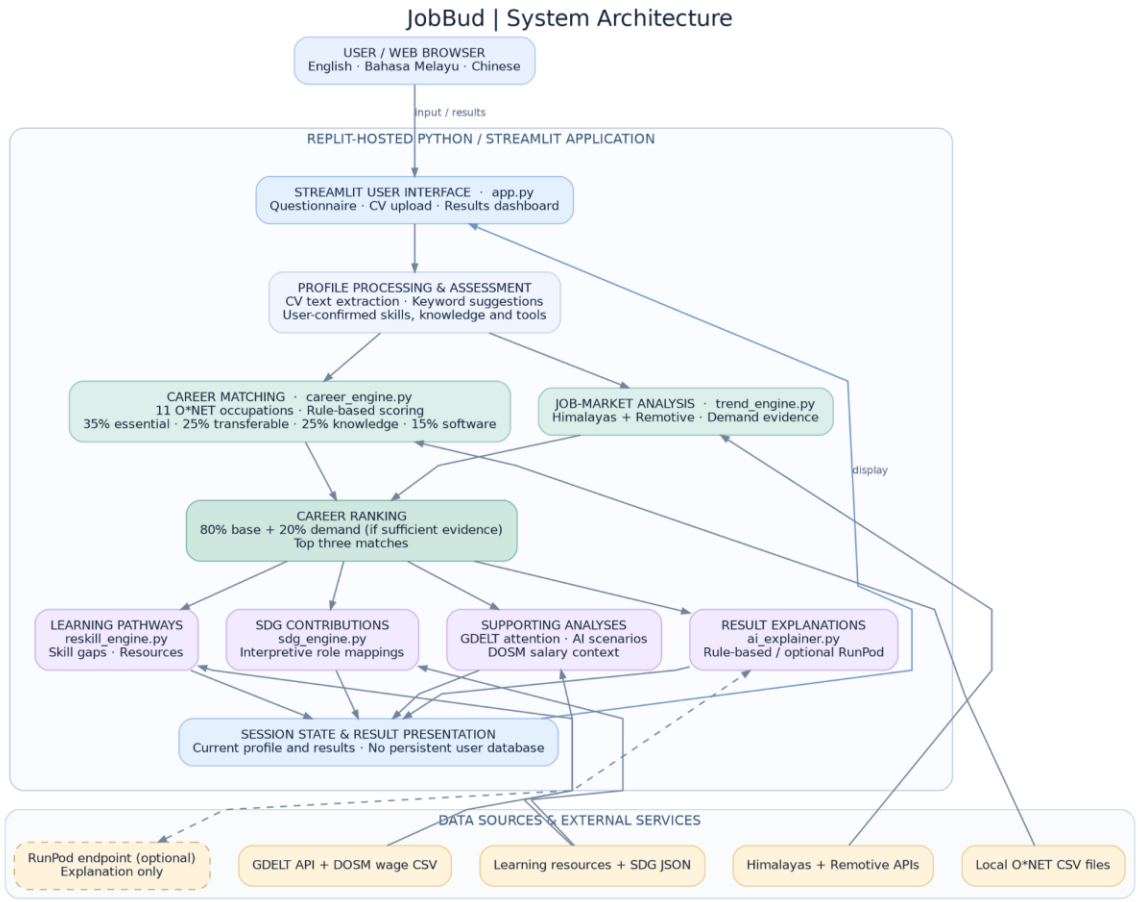
List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

To build a functional, lightweight, and responsive prototype, our solution utilizes the following stack:

- Frontend & Web Framework: Streamlit (Python) for building interactive data dashboards, user profile intake forms, and dynamic visualization tabs.
- Backend & Data Processing Engine: Python combined with Pandas and NumPy to process structured datasets, execute O*NET skills-matching algorithms, and compute alignment and readiness scores.
- Database & Data Sources: Structured occupational datasets from O*NET, including occupation data, essential skills, transferable skills, knowledge, workplace software, and related occupations. Additional external sources include Himalayas and Remotive for current remote job-posting signals, GDELT for AI and technology news-attention signals, OpenDOSM for Malaysian industry-level salary context, team-curated learning resources, and UN SDG references.
- Deployment & Hosting Environment: Replit for cloud-based development, environment management, and live web app hosting.

Below is the system architecture illustrating how data flows from user input to the final deployed interface:



Platform Utilized: Replit

We use Replit as the primary cloud-based development, testing, and deployment environment for JobBud. Replit provides an integrated Python workspace, package and dependency management, collaborative development tools, and browser-accessible application hosting. The platform enables our team to prototype, test, debug, and deploy the Streamlit application efficiently while maintaining an accessible proof-of-concept for evaluation.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied the design-thinking methodology to ensure our solution is technically viable, user-friendly, and ready for Phase 2 deployment. By centering our research on fresh graduates and career switchers navigating analytical and financial fields, we defined the primary challenge: eliminating career friction caused by rapid AI disruption through transparent, data-driven guidance. During the ideation phase, we mapped out a unified workflow incorporating all six must-have features, including dynamic skill profile intake, updatable trend signals, explainable role matching, and UN SDG alignment. To guarantee technical viability and guard against AI hallucinations, we engineered a deterministic Python-backed processing engine using Pandas to analyze structured O*NET datasets (occupation_data.csv, essential_skills.csv, etc.). Building on insights from the SDC Phase 1 workshops, we prototyped and deployed our cloud-hosted Streamlit application on Replit. This lightweight architecture ensures instant, accurate gap analysis and custom reskilling pathways while keeping human judgment at the center of career decisions. By prioritizing data integrity, modularity, and an intuitive UI, our prototype is ready for Phase 2 testing, validation, and further development toward real-world deployment.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Target 4.4**: Substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs, and entrepreneurship. **Target 8.5**: Achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	TalentCorp's Impact Study on AI, Digital and the Green Economy estimates that approximately 620,000 Malaysian employees, representing about 18% of the workforce across the sectors analysed, are expected to be significantly affected by technological and economic transformation over the next three to five years. This highlights the need for accessible skills assessment, reskilling guidance, and clearer career-transition pathways. Source: Talent Corporation Malaysia Berhad, Impact Study of AI, Digital and Green Economy on the Malaysian Workforce.
Core Measurable KPI *The specific unit you will track*	1. Career Skill Alignment Score (%): the weighted alignment between a participant's assessed profile and O*NET occupational requirements.

Metric Category	Your Response
e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.	2. Identified Skill Gap Resolution Rate (%): the proportion of priority development gaps for which the participant completes or begins a recommended learning action during the test period. 3. Assessment Coverage (%): the proportion of relevant occupational evidence covered by the participant's completed assessment.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	For the Phase 2 test cohort, JobBud will establish a pre-assessment baseline and compare it with a post-guidance assessment. The target is to achieve at least a 20% improvement in users' understanding of their priority career-skill gaps and recommended next steps, measured through a short pre/post evaluation questionnaire, while tracking changes in skill-alignment indicators separately rather than claiming that the application itself directly increases users' skills.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

[ ] Yes [/] No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

[/] Yes [ ] No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
occupation_data.csv	Dataset	CC BY 4.0
essential_skills.csv	Dataset	CC BY 4.0
transferable_skills.csv	Dataset	CC BY 4.0
knowledge.csv	Dataset	CC BY 4.0
software_skills.csv	Dataset	CC BY 4.0
Remotive Job Listings	Dataset	Provider API terms; attribution required
Himalayas Job Listings	Datasset	Provider API terms; attribution required
GDELT News Data	Dataset	Open use; attribution required

Malaysian Industry Salary Data (wages_industry.csv)	Dataset	CC BY 4.0
learning_resources.json	Dataset	Team-curated; underlying sources retain their terms
User Career Profiles and Skill Assessments	Dataset	User-provided data; subject to applicable privacy requirements
Uploaded CVs	Dataset	User-provided data; subject to applicable privacy requirements

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

[ ] Yes [/] No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Signature] PO ROU JING

[Date] 24/10/2026